slopninja

Private writing models, adversarial evaluation, and paid inference on Bittensor

slopninja research

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Abstract

slopninja proposes a Bittensor subnet for author and AI-origin detection, author-directed text transformation, and writing improvement. Miners develop private models and compete on the usefulness of their outputs. They receive network emissions for benchmark performance and can separately sell asynchronous inference at market prices denominated in the subnet's alpha token. The protocol requires no publication of model weights, feature definitions, adapters, or training recipes.

Every customer inference request is encrypted for its assigned miner. The customer's source text, reference writing, style profile, and editorial brief are not automatically disclosed to validators, routing services, or other miners. Customers receive encrypted results and may separately provide evidence to a specific validator of their choice. Subnet-owned benchmarks provide the authorized material that validators need to assess accuracy, preservation, style, and writing quality. Transformation miners compete against both Pangram and the subnet's best eligible AI-origin detector. Retired benchmark rewrites feed later detection rounds, creating an adversarial training loop. Detector reports establish observations rather than human authorship or preserved meaning.

On-chain contracts coordinate offers, commitments, deadlines, and payments; miners execute inference privately off-chain. Public randomness determines benchmark audits after submissions are committed. This draft specifies the proposed architecture, training data, reward rules, and trust assumptions. It also reports the limits of the existing prototype. A reliable rewriting model, private inference service, and live reward mechanism remain to be demonstrated.

1 Purpose and scope

A useful writing model should retain a writer's argument while allowing changes to expression. The same person may want a passage closer to their own voice, more accessible to a general audience, or more forceful without becoming less accurate. An authorship model can help describe those differences. A detector can measure one aspect of how the result is classified. Neither measurement alone establishes that the revision is good.

The subnet therefore evaluates three tasks separately:

1. Identify authors and documented human/model production histories.
2. Transform writing toward a requested voice, optionally reducing detection by Pangram and the subnet's own best AI detection model, while preserving the source's meaning and intended tone.
3. Improve writing for an explicit audience and editorial purpose.

The joint evasion target requires Pangram's reported AI-generated **plus AI-assisted** fraction strictly below 10% and a pass against the subnet detector at its separately calibrated threshold, with preservation and quality requirements satisfied. Reaching zero is a desirable measured outcome, not a promise of this design. A model-origin text retains that recorded origin after a successful rewrite. Detector evasion does not change its production history.

The adversarial loop connects the first two tasks: detection miners learn to recognize transformed model-origin text, while transformation miners try to defeat the best eligible detector from a completed round. The benchmark fixes that opponent before accepting rewrites and retains the source's production labels as the models improve.

The initial scope is English prose, document-level author/origin predictions, and bounded asynchronous editing jobs. Span-level origin labels, multilingual rewards, and arbitrary-length manuscripts need their own validated datasets. There is no requirement for token streaming or interactive latency.

2 Related subnets, services, and research

This representative survey covers direct task competitors and relevant evaluation and inference infrastructure. It uses project-maintained documentation and research publications checked on September 9, 2026. Subnet numbers below follow the projects' own documentation; this is not an independent audit of live chain state. Published service claims are distinguished from results measured by this project.

2.1 Bittensor precedents

It's AI and BitMind's GAS supply the closest task and incentive precedents. It's AI documents human/AI text classification on SN32, with miners serving detection requests. Its FAQ says the team does not know which models its leading miners use. This provides a precedent for evaluating detection endpoints with undisclosed implementations. GAS documents an explicit competition between detection models and generators that try to fool them [1–3].

Project	Documented approach	Relevance to slopninja
It's AI (SN32)	Human/AI text detection, including sentence-level probabilities, through miner endpoints [1,2]	Direct comparison for origin detection; compare calibration and generalization on the same independently labeled text
BitMind GAS (SN34)	Image/video/audio detector submissions and image/video generator endpoints; generation rewards include fooling detectors [3,4]	Direct precedent for adversarial incentives; text revision adds source-preservation and requested-voice constraints
Macrocosmos Finetuning (SN37)	Miners publish competition models on public Hugging Face repositories for validator evaluation [5]	Precedent for rewarding model improvements through submitted weights; slopninja proposes private model serving
Chutes (SN64)	Serverless model inference and metered service, with documented confidential-serving options [6,7]	Paid inference and private serving infrastructure; writing quality requires separate task evaluation
Targon (SN4)	Managed compute and inference; confidential VMs with attestation-bound key release for proprietary workloads [8,9]	Hardware-based endpoint protection and private-model execution are established design topics

These projects establish that detection competition, adversarial generation, language-model training rewards, and paid private inference already have Bittensor precedents. Their documented mechanisms evaluate different properties. For example, GAS requires submitted detector weights, whereas slopninja permits private detector services. Chutes documents confidential serving and request/response encryption, and Targon describes hardware attestation and key release. Those hardware assumptions differ from this paper's default trust in the assigned miner after decryption. A secure execution environment alone does not establish that a revision preserved an argument.

2.2 Existing writing services

Pangram and GPTZero already offer human/model-origin classification, including mixed or assisted categories. Pangram is the required external opponent here; GPTZero is another useful comparison on rights-cleared benchmark material. Their output categories and scores must be mapped explicitly to the evaluation's documented production labels. Agreement among detectors does not establish an author's identity [10,11].

Grammarly documents personal voice profiles and a feature for rewriting in the user's voice. Wordtune offers sentence and paragraph rewriting, formality changes, shortening, and expansion. QuillBot documents multiple paraphrasing modes, including custom and Humanizer modes, and warns that its Creative mode can alter meaning. These are direct comparators for author-directed revision and writing improvement; personalization and editorial controls are existing product capabilities [12–14].

Undetectable AI explicitly markets humanization modes intended to bypass detectors. That documents an existing commercial objective; this survey does not independently establish its bypass or preservation performance [15]. Comparing such services requires the same sources, briefs, retained outputs, and blinded preservation review used for subnet miners. A product's advertised detector score cannot substitute for that comparison. Hosted service comparisons use authorized benchmark text and do not create an exception to confidential customer delivery.

2.3 Research foundations and proposed contribution

Gero et al. model style through controls over function-word frequencies and syntactic constructions while retaining content words. This is a direct antecedent for combining lexical and grammatical features. StyleRemix uses interpretable style axes and LoRA modules for authorship obfuscation, providing a relevant comparison for controllable rewriting [16,17]. Authorship obfuscation changes evidence available to an author classifier; matching a requested writer and preserving every required claim remain separate evaluation questions.

The PAN/ELOQUENT 2024 task paired human/AI detection with generation and obfuscation intended to avoid it, offering a prior benchmark for an adversarial loop. DIPPER demonstrated paraphrase attacks on several then-current detectors and studied a retrieval-based defense. Those results motivate adaptive opponents and held-out evaluation; they do not establish performance against Pangram 4 or slopninja's future incumbents [18,19].

The proposed contribution is a text-focused protocol connecting origin-detection competition to author-directed transformation, with independent preservation review and a separate writing-improvement reward. Transformation miners face both Pangram and the best eligible subnet detector; retired, authorized rewrites train and test later detectors. Miners retain their models and can sell jobs through an alpha-denominated market with explicit recipient encryption. This combination requires validation against the precedents above. The survey establishes neither uniqueness across all subnets nor superior performance by an implementation that has not yet been built.

3 Participants and the two kinds of work

Miners register service offers and retain their models. Customers buy specific jobs from those offers. Validators evaluate subnet-owned benchmark tasks and submit weights. Benchmark curators maintain permitted source material, production records, and hidden evaluation labels. A customer-facing gateway may relay ciphertext, but receives no decryption key.

Property	Customer inference	Subnet benchmark
Purpose	Serve a customer's request	Measure miners for emissions
Source of text	Customer	Authorized benchmark collection
Default input recipients	Customer and assigned miner	Task owner, assigned miner, authorized evaluators
Evaluation	Customer acceptance	Fixed scoring policy and independent review
Pangram submission	Excluded from the confidential job protocol	Required for evasion tasks
Payment	Customer's quoted fee	Performance-dependent emissions
Training reuse	No automatic reuse	Only under the collection's release terms

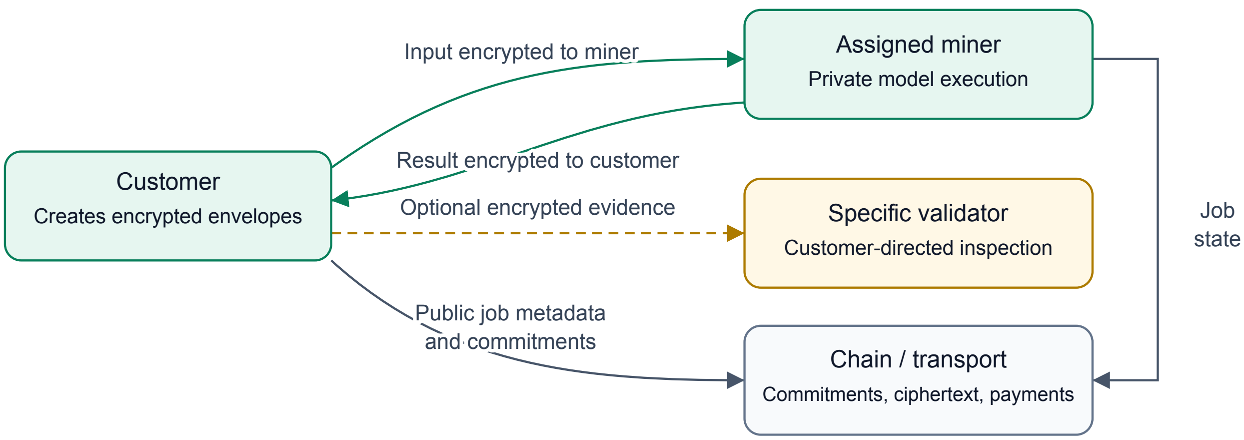

Figure 1: Customer inference and optional evidence disclosure. Only the addressed recipient can decrypt each envelope; endpoint handling remains a trust assumption.

Customer tasks do not become benchmark examples automatically. Validators must not obtain customer inputs through an audit exception, a shared support key, or an arbitration process. A customer may initiate a separate, explicitly addressed evidence disclosure as described in Section 6.3. This applies to detection requests as well as both editing tasks. An author's uploaded reference collection can be more identifying than the query itself and receives the same protection.

Miners may advertise specialties, prices, length limits, and delivery times. The network evaluates their observable service performance. It cannot infer how many GPU hours they used, whether they own every component, or which private weights produced a response. Emissions create an incentive to improve training; they do not reimburse unverifiable training claims.

4 Model interfaces and starter architectures

The three interfaces define compatible jobs and outputs. Miners may change their internal architectures, grammatical representations, search methods, and training data without changing those interfaces. Sharing a foundation model between tasks is permitted. Publishing a winning model is optional.

4.1 Author and origin detection

An author task supplies a query and reference works for a gallery of candidate authors. The response contains probabilities over the supplied author IDs. The starter compares full sparse word/grammar profiles; a learned text encoder with an author embedding head is the next architectural comparison. Reference prototypes support new customer authors without fitting a permanent classifier for each identity. Unknown-author rejection requires a separate calibrated output and withheld-author evaluation.

An origin task returns probabilities over documented production histories: human-only, model-only, and mixed human/model work. The production record includes revision order and assistance type. A model-to-model rewrite remains model-only under this initial label scheme. Pangram supplies a comparison, not the training label. The proposed encoder uses a separate origin head; author similarity must not be substituted for provenance.

ModernBERT-base [20], a 149M encoder, is a concrete starter candidate. Selection must compare it with the existing metric on new authors, topics, and registers. The subnet does not require that checkpoint or assume it will win.

4.2 Author transformation and detector evasion

Inputs are the source, independent target-author samples, requested style changes, and a preservation brief. The task declares author matching, evasion against both Pangram and the selected subnet AI-origin detector, or both objectives. The miner returns exact revised text or an abstention. Opponent selection and scoring are fixed by the benchmark, independently of the submitting miner.

The proposed starter is an 8B decoder, initially Qwen3-8B [21], with a style adapter and a candidate-selection process. Named word and grammar deviations can condition the first version. A learned profile prefix is a later comparison, contingent on better results for unseen authors. Deterministic Rust edit rules can propose small changes; the decoder can propose broader restructuring.

Train first on reviewed source/reference/brief/revision tuples. Then learn preferences among candidates that pass preservation review. Include originals, failed rewrites, copied reference passages, and fluent meaning-breaking edits as controls. A miner's own style score may guide its search, but cannot approve its output for benchmark rewards.

4.3 Writing improvement

This task supplies an audience, purpose, tone, and allowed changes. The miner returns an improved revision or leaves suitable text unchanged. The starter shares the decoder base with transformation and uses a separate editorial adapter and preference scorer when the tasks' requested behavior conflicts.

Training preferences distinguish clarity, accessibility, organization, rhetorical force, and tone. A lower reading level or shorter text is useful only when the brief calls for it. Reviewer ties and disagreement remain in the data. Validators use independent judges and reader audits; a miner cannot determine its own payment by scoring its revision.

Pangram is not part of the writing-improvement reward. Future comparisons of 4B, 8B, and 32B editing models should use fidelity, reader preference, and cost per accepted revision. All checkpoint licenses and revisions must be pinned before a training release. The named starters are proposals, not trained subnet products.

5 Training data and evaluation separation

The data program needs three linked tables, with different supervision:

Table	Records	Supervision
Authorship	Multiple independent works per writer	Identity, date, genre, work family, provenance strength
Production	Original writing, generations, and revision chains	Recorded workflow, parent links, generator version, assistance order
Revision preferences	Sources, briefs, candidate edits, and reviews	Preservation, target voice, editorial preference, ties

Maintain distinct training, public development, private evaluation, and customer reference collections. Split authors before selecting vocabulary or fitting weights. Within evaluation authors, use independent earlier works as references and later works as queries. Keep every revision, translation, and excerpt of one source family in the same split. Hold out generator families and prompt families for origin evaluation. Match content, length, and genre across origins so that a detector cannot win by identifying the assignment format.

Global Voices supplies a prospective pool of attributed modern prose under its default CC BY terms. Qualification must distinguish original writers, translators, shared accounts, and third-party material. PLOS articles can supply factual source material and citation constraints, while their coauthored nature makes them unsuitable as unqualified single-writer labels. Item-level rights and attribution remain part of acquisition. Global Voices policy [22], PLOS article policy [23].

Historical publication alone does not prove a human-only workflow. For stronger labels and useful editing supervision, the proposed pilot commissions 50 writers to provide twelve independent pieces each and two edits of supplied drafts. Common briefs across writers permit comparisons with similar subject matter; multiple subjects per writer test consistency beyond topic. Record collaboration and AI assistance, and obtain separate rights for training, private evaluation, and any redistribution.

If collection and review prove workable, expand to 500 writers: 300 training, 100 development, and 100 private evaluation authors. A further cohort is needed for unknown-author rejection and stronger confirmation. These are acquisition targets; the writers have not been commissioned and this paper commits no funds.

Generate model-only and mixed examples from permitted sources while retaining exact prompts, model revisions, outputs, and human editing records. Synthetic restoration pairs can start from a writer's original, create a generic draft, and train a return toward the original. Review whether that draft still contains the required information; otherwise restoration teaches invention. Keep such weak supervision separate from real writer edits.

Rewrites that fool Pangram or the selected subnet detector are hard examples for later detection rounds. Adversarial benchmark revisions retain their original production labels. They may become later detector-training examples only after retirement from private evaluation and under agreed release terms. Preserve difficult human examples alongside them. Customer text, profiles, and outputs receive no automatic training or benchmark reuse.

The existing Blog Authorship Corpus experiments remain local noncommercial research and do not supply the commercial starter dataset. Private book and correspondence material is also excluded. Source manifests must distinguish text rights from loader-code licenses. The detailed acquisition plan [24] records the current source screening and outstanding qualification work.

6 Confidential customer inference

6.1 Assigned-miner encryption

For every customer job, the protocol gives input-decryption capability only to its assigned miner. The customer already possesses the input and may retain it. Validators, gateways, storage operators, other miners, and chain observers receive no input key or alternate recipient envelope. Models execute on the assigned miner's infrastructure; confidential jobs may not forward plaintext to an external model API, detector, or another miner.

A miner advertises a separate X25519 encryption key in a hotkey-signed record binding the network, service, key ID, validity window, and job or offer scope. Signing and encryption keys have distinct roles. The customer verifies the miner identity and finalized key registration, reserves that miner's offer, and encrypts locally before sending any payload to the network. A gateway cannot select a substitute key after receiving plaintext.

The proposed HPKE suite uses X25519 with HKDF-SHA256 for key encapsulation, HKDF-SHA256 for key derivation, and ChaCha20Poly1305 for authenticated encryption. A customer signature authenticates the envelope and its job context; HPKE Base does not itself authenticate the sender. Bind the chain, subnet, contract, job ID, assignment generation, offer hash, recipient key ID, expiry, and ciphertext to the authenticated request. Check replay state and assignment validity before processing. These are application requirements around the HPKE primitive. RFC 9180 [25].

The encrypted payload includes the source, reference works, style profile, brief, private metadata, and a commitment opening. Public commitments use a versioned, unambiguous encoding and a fresh secret 32-byte salt. An illustrative commitment is

$$C = \mathrm{SHA256}\big(\mathrm{domain} \parallel \mathrm{context} \parallel \mathrm{salt} \parallel \mathrm{length} \parallel \mathrm{payload}\big). \tag{1}$$

Plain text hashes, filenames, report URLs, and descriptive task titles stay inside the envelope. Padding limits exact length leakage; public size buckets, timing, prices, and counterparties remain observable.

The result is encrypted to the customer's authenticated output key. The miner knows the result it generated, but supplies no result envelope to validators. The same rule covers probabilities, explanations, and change notes: detection results can reveal sensitive inputs too.

6.2 Assignment, recovery, and the privacy boundary

Each assignment binds one miner and one encryption-key version. Rotation cannot silently redirect an existing request. A retry with another miner requires the customer to authorize a new assignment and encrypt a new request. The protocol does not escrow recovery keys or re-encrypt inputs for a fallback miner. If the customer is unavailable, a failed job expires under its settlement terms. Reassignment cannot erase a previous miner's knowledge of text it already read.

Use short-lived, preferably job-scoped keys and an explicit retention policy. Ordinary recipient encryption does not establish that a miner deleted keys, plaintext, or logs. A compromised recipient key may expose recorded ciphertext. An assigned miner can copy plaintext after decrypting it; this design protects against unauthorized protocol recipients and observers, and assumes the chosen miner follows its data-handling terms. It does not prove secrecy against that authorized endpoint. Hardware attestation or computation over encrypted data would require a separately specified and evaluated design.

There is no automatic exception for customer support, quality disputes, subnet spot checks, or Pangram verification. A customer may separately choose to disclose material to another party. In particular, the subnet must not advertise a public Pangram report as compatible with exclusive miner access.

6.3 Customer-directed evidence disclosure

A customer can ask a specific validator to inspect a job by creating a new evidence packet encrypted to that validator's authenticated key. The customer signs a disclosure context binding the job, selected validator and key version, purpose, scope, and validity window. The packet can contain source passages, the delivered result, the miner's signed response, and relevant commitment openings. The customer selects what to send. Neither the miner nor a gateway can activate this access on the customer's behalf.

This grants the selected validator access to that packet, with no decryption capability for other jobs or undisclosed material. The validator must not forward it to other reviewers or providers without further customer direction. Expiry ends authorization for new processing; it cannot erase plaintext the recipient already obtained. Keep the packet, findings, and openings private; publish only any agreed decision or commitment needed by the protocol.

The validator checks which supplied evidence opens the original commitments. A redacted excerpt does not open a commitment to a complete document. Full openings or a separately specified selective-disclosure commitment scheme are needed for stronger binding. Incomplete evidence can support a limited opinion, but cannot establish fidelity of the whole job.

Inspection is advisory by default. A signed decision affects escrow only if the job's original terms gave that validator or selection policy authority and both parties accepted those terms. Choosing a reviewer does not itself change payment rights or extend deadlines. This route permits voluntary inspection while preserving the default assigned-miner input boundary.

7 Paid asynchronous jobs and on-chain coordination

The proposed job contract uses Subtensor's EVM to coordinate offers, escrow, commitments, and settlement. Inference prices are denominated in slopninja's subnet alpha; actual model execution stays on miner hardware. Subtensor EVM guide [26].

7.1 Market pricing in alpha

Miners publish signed offers for bounded jobs. Each offer specifies a total alpha price, task and service version, input/output limits, delivery deadline, available capacity, quote expiry, and settlement terms. Customers select among compatible offers using price, benchmark performance, and delivery time. A higher-quality model or faster service may command a different price. There need not be one clearing price for every kind of inference.

Miners may revise their unreserved offers as demand, queue length, compute costs, and competition change. The subnet operator sets no base price, utilization target, or global price-adjustment coefficient. Miners choose their own pricing policies; customers decide whether the offers are worth buying. Alpha is the unit of account and proposed settlement asset. The inference market determines the number of alpha units charged independently of the alpha/TAO exchange market.

An atomic reservation consumes offered capacity and locks the quoted alpha amount and job terms. Later price changes apply only to later reservations. An expired or exhausted offer cannot be filled; repricing requires a new customer choice. The reservation therefore gives both parties a known price through execution and settlement, even while other offers change.

Quote discovery uses public service descriptions and the size/deadline information the customer chooses to reveal. Source text, reference writing, and the full brief remain encrypted for the selected miner. Obtaining competing offers does not distribute the plaintext to prospective bidders. If the selected miner cannot serve the bounded request, the job follows its rejection/refund terms; it cannot silently raise the price after decrypting the input.

7.2 Reservation and delivery

Subtensor's staking-v2 interface exposes allowances and stake transfers, and contracts can hold stake through their mapped coldkeys. The proposed escrow adapter would use those operations to custody unlocked, transferable alpha on the same subnet. An allowance alone does not fund a reservation. Alpha remains associated with a subnet and hotkey; settlement must account for that position and its native 10^{-9} precision. These primitives do not supply a finished job-escrow implementation [27,28].

An offer distinguishes the service price from network and transfer fees and names the fee payer. Funding succeeds only after the credited alpha principal satisfies the quote. Release and refund must reconcile actual balances, transfer permissions, and fees; incidental stake accrual must remain separate from job principal. The adapter and its accounting policy require validation before paid launch. Neither an alpha price feed nor an assumed ERC-20 interface substitutes for that work.

The proposed lifecycle is:

1. **Offer and reserve.** The customer selects a miner's signed offer, pins its key and job terms, and escrows the quoted alpha amount for a bounded input/output.

2. **Encrypt and submit.** The customer posts the salted input commitment and encrypted-delivery hash, then delivers the envelope to that miner.
3. **Accept.** The miner decrypts, checks the opening and limits, and accepts before the acceptance deadline. Failure to accept permits a refund.
4. **Execute and deliver.** The miner runs the private service, commits the result, and makes its customer-encrypted envelope available before the delivery deadline.
5. **Acknowledge or expire.** A customer acknowledgment releases payment. No timely acceptance or delivery claim permits refund. A delivery claim without acknowledgment follows the bounded timeout policy below.

Bind acknowledgments and state transitions to chain, contract, job ID, role, nonce, and assignment generation. Bind the alpha asset's subnet ID and amount in native units as well. Authenticate the association between a Bittensor hotkey and its settlement identity. A hash or truncated address is not an authorization method. Enforce exactly one terminal settlement and explicit finality rules. A customer, miner, or keeper must submit timeout transactions; deadlines do not execute themselves.

Customers may deliver ciphertext through authenticated asynchronous endpoints and redundant encrypted storage. Small envelopes could instead travel through on-chain calldata or events, so miners can discover and complete jobs entirely through chain messaging. Payload-size, gas, and retention measurements must precede that choice. Storing ciphertext on-chain does not hide metadata or provide permanent secrecy after key compromise.

7.3 Payment without disclosing the job

A customer can consume a useful result and refuse acknowledgment. A miner can claim delivery of useless or undecryptable bytes. Public commitments alone do not distinguish those cases. Validators who cannot read the job also cannot adjudicate its meaning or writing quality.

For the first private-job pilot, use customer acknowledgment for payment and a fixed, bounded refund timeout for unacknowledged jobs. Publish that policy in the offer before acceptance. This assigns nonpayment risk to the miner, including when a customer has already consumed a valid result. Cap pilot job values and record the loss rate. It is an explicit service tradeoff, not trustless fair exchange. Public protocol evidence can resolve duplicate or late transactions; it cannot resolve a secret editorial disagreement.

Benchmark reputation can help customers choose a service without exposing their own work. It does not prove any particular unknown-input prediction correct. The customer-directed inspection route in Section 6.3 can support an agreed dispute policy. Its effect on payment must be fixed before job acceptance; a customer-selected opinion alone cannot redirect escrow. The initial pilot retains acknowledgment and bounded timeout refunds. Stronger private payment guarantees remain a launch research question.

7.4 Revenue and costs

Customer fees pay for the purchased service. Emissions reward performance on separate benchmark assignments. Customer volume does not directly increase emission rewards, because miners can transact with themselves. A fixed quote avoids charging for hidden, unverifiable internal token or search counts.

Miners pay their own training, inference, private search, and required candidate Pangram submissions for benchmarks. The benchmark operator or participating validators fund source-baseline reports from an explicit round budget; those reports can be reused across competing miners under the same task/version. Validators also bear retrieval, hosting, and review costs. Subsidies, if used, must identify their funder and cap before assignments begin. This design assumes no free API-credit allocation or automatic on-chain reimbursement scheme.

Required calls to the selected subnet detector also need an explicit round budget and service terms with that miner. The benchmark operator or participating validators fund those calls; miners fund any additional private search. This access applies to authorized benchmark text. Confidential customer jobs cannot automatically forward inputs or outputs to Pangram or the selected detector's operator.

8 Benchmark scoring and mechanism mapping

The current Bittensor documentation permits two chain mechanisms, each with separate weights and consensus while participants share UIDs. slopninja maps author/origin detection to mechanism 0 and combines separately scored transformation and writing-improvement pools in mechanism 1. The three logical tasks do not require three chain mechanisms. Deployment must also respect the documented joint UID/mechanism capacity constraint. Bittensor mechanisms [29].

For an initial offline tournament, give the three logical tasks equal budget shares. Within detection, balance author and origin tasks equally; within transformation, retain separate author-style, evasion, and combined modes. These are transparent pilot allocations, not an economically optimized split. Any live allocation and change policy must be published before the scoring epoch. Miners may specialize and earn from the pools they enter.

8.1 Detection

For a valid probability vector p and known one-hot label y, use the multiclass Brier utility:

$$B(p, y) = 1 - \frac{1}{2}\sum_{k}(p_k - y_k)^2. \tag{2}$$

Average query results within source groups and authors before the declared task mixture. Compare the aggregate with a fixed public baseline, then use positive improvement divided by that baseline's remaining headroom. If the baseline is perfect, the pool has no remaining improvement and is unallocated in the offline simulation. Clip after aggregate comparison, not separately for each lucky prediction. Raw Brier utility is unsuitable for direct payout normalization: a uniform prediction in a large author gallery can already have a high value.

Report retrieval accuracy, reciprocal rank, origin calibration, and human false-positive rates separately. Fix operating thresholds on development data. Unknown or weak production records cannot become hard labels merely because Pangram agrees with a miner.

8.2 The adversarial feedback loop

Before each transformation round, select the strongest qualified AI-origin service from a completed detection round. Rank eligible services by held-out origin Brier performance, subject to the declared calibration and human false-positive requirements. Author-retrieval scores do not select this opponent. Freeze its identity, service version, threshold, request settings, and response policy before releasing transformation assignments. Bootstrap rounds use a named reference detector until a competitive incumbent qualifies and are reported separately.

The selected service cannot score its own operator's transformation submissions for rewards. Such assignments use the highest-ranked eligible independent opponent under a rule fixed before the round. Retain the opponent identity with every score and compare miners on matched assignments. Declared independence, signed responses, blinded control requests, and consistency checks reduce conflicts; they cannot prove that undisclosed collusion or selective service behavior is absent. A private version label does not attest which weights ran. Unavailable or invalid opponent evidence follows the frozen failure policy and cannot turn into a Pangram-only joint pass.

Transformation miners try to defeat this opponent and Pangram while passing independent preservation and style review. Authorized benchmark rewrites then become hard examples for later detection rounds, with their production labels retained. Split complete source families between released training material and private evaluation; release training examples only after retiring them from evaluation. Maintain hard human examples alongside generated rewrites. Neither immediate publication of test answers nor customer-text reuse is part of this feedback loop.

8.3 Preservation before revision rewards

Each editing task has a preservation brief and hidden review material covering claims, participants, quantities, negation, uncertainty, citations, attribution, and argument relationships. Review the entire revision. Independent judges also check readability and the requested tone. A confirmed failure sets the task's preservation gate G to zero; a confirmed pass sets it to one.

Readers compare anonymized revisions in randomized order without detector or miner scores. Maintain unchanged sources and qualified editor baselines. Automated judges join reward allocation only after agreement and failure modes have been measured against separate reader judgments. Instructions embedded in a candidate are evaluated text, not evaluator instructions.

An unresolved judgment cannot pass by default. The round policy must resolve it or void the affected comparison for all competing miners consistently. Miner nonresponse remains a zero on assigned work; validator infrastructure failure is a separate, recorded event. Publish coverage and missingness so that conditional accuracy cannot conceal excluded difficult cases.

8.4 Transformation and improvement utility

Let $A \in [0, 1]$ be normalized positive improvement in target-style fit over the unchanged source. Estimate it from independent author models and blinded target-author or reader comparisons, retaining uncertainty. Before using a particular combination for live rewards, freeze and validate it on development examples. Topic copying, reference copying, and meaning-breaking edits must fail that validation. No miner-defined embedding can be its own reward oracle.

For Pangram evasion, let $F_r(t)$ be the reported AI-plus-assisted fraction in required observation r for text t. The initial benchmark requires three distinct completed reports for each source x and candidate z. Improvement using the least favorable pair among those submitted observations is:

$$\Delta_P(x, z) = \min_r F_r(x) - \max_r F_r(z). \tag{3}$$

The strict target additionally requires $\max_r F_r(z) < 0.10$. Three reads of one report count as one observation. A neighborhood such as two percentage points does not relax the strict target. Use the declared provider version and repeat policy for both sides; report sensitivity to provider variation.

Miners may pay for additional search and report attempts before commitment. Scoring uses the committed reports, including variation in their provider results. The protocol does not require disclosure of every private attempt or a fresh validator-paid Pangram inference call. A reported pass applies to the committed observations.

For the selected subnet detector, let $H(t) = p(\text{model-only} \mid t) + p(\text{mixed} \mid t)$, using its required, authenticated response under the frozen policy. This is a document-origin probability, while Pangram's F_r is a content fraction. Define $\Delta_S(x, z) = H(x) - H(z)$ and report the two detector improvements separately. The pilot joint evasion utility is

$$E(x, z) = \begin{cases} \big(\Delta_P(x, z) + \Delta_S(x, z)\big)/2, & \Delta_P(x, z) \geq 0 \text{ and } \Delta_S(x, z) \geq 0, \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

The joint pass requires both $\max_r F_r(z) < 0.10$ and $H(z) < \tau_S$, where τ_S is calibrated on independent development data at the declared human false-positive operating point and frozen before the round. It is not automatically 0.10: the two detectors report different quantities. Neither detector's improvement compensates for regression against the other.

Pilot transformation utilities are GA, GE, and $G(A + E)/2$ for the three modes. Combined mode also requires no regression on either requested axis. Report sub-10% success separately from incremental improvement. An already suitable source can stay unchanged without receiving an invented improvement bonus.

For writing improvement, use positive blinded preference improvement over the fixed qualified editor baseline, subject to preservation and no regression against the original. Keep ties. Neither AI detector contributes a term to this utility. The first tournament determines whether these proposed utilities order useful revisions above omissions, unsupported strengthening, irrelevant padding, and needless edits.

8.5 Aggregation and failure policy

Fix assignments, task mixtures, baselines, and deadlines before submissions. Balance related examples within source groups; extra cheap submissions do not increase a miner's share. Normalize positive skill within each task pool, then combine pool distributions using the published shares. The chain mechanism split carries the corresponding aggregate task allocation.

If no miner beats a baseline, record that pool as unallocated in the offline simulation. A live adapter must select and test a chain-valid fallback before launch. This paper does not assume that zero weights, retained emissions, a new burn destination, or custom slashing are available. Payout handling under empty pools and unresolved rounds is a required deployment decision.

9 Detector evidence and randomly assigned audits

Pangram's API provides asynchronous tasks, analyzed text, document fractions, version information, and optional public dashboard links. The returned text may differ through preprocessing, so validators must bind evidence to the analyzed bytes rather than a preview. The local reader currently requires exact equality; a normalization policy would need its own version and review. Pangram API [30].

Our synthetic probe obtained a publicly retrievable report containing the analyzed text and scores. An independent caller retrieved it without a new paid inference. The observed web endpoint is undocumented and needs an availability, retention, and usage agreement before production dependence. We have not established a provider signature, immutable model identity, or one unique canonical result per text. Recorded probe [31].

For an evasion benchmark, a miner commits its final candidate and required report IDs before audit selection. Assigned validators receive the source, candidate, report URLs, score projections, and commitment openings in separate encrypted envelopes. They retrieve the provider's record directly and compare the exact text, fractions, version, and time window. Report IDs and URLs remain out of public logs and chain payloads. Anyone who obtains a public URL may still retrieve its content; encrypting the locator protects our delivery path, not Pangram's hosting.

Pangram states that it does not train Pangram 4 on customer API data. Private miner weights prevent direct downloading, but do not prevent independent black-box study through purchased outputs or leaked examples. The subnet must not assert otherwise. Provider verification and any reuse of provider labels also need suitable service terms. Pangram model card [10].

For the selected subnet detector, retain signed responses binding the round, frozen service identity and settings, request nonce, text commitment, and full origin-probability vector. The benchmark coordinator obtains the required candidate response after the candidate commitment; private search results cannot replace it. Record that evidence before the audit beacon, then deliver its openings only to authorized auditors. They verify the signer and bindings and check the declared control and consistency policy. A signature establishes the service's response; it does not prove execution of particular secret weights. The protocol must validate this evidence path independently of Pangram's public-report reader. Customer requests never enter it automatically.

9.1 Audit sequence

1. Freeze the benchmark policy, authorized auditor roster, question-pool root, selected subnet detector, and source-baseline evidence. Questions and answers stay private; public roots use salted commitments.
2. Accept one final submission commitment per assignment. It binds candidate text, reports, task identity, and the offered service version. Finalize all eligible commitments before a predetermined future drand round, with an explicit timing margin. Obtain and commit the required subnet-detector response before that round as well.
3. Verify the pinned beacon's chain, public key, scheme, round, and signature. Derive a domain-separated seed from that beacon and the frozen epoch/pool commitments. Use unbiased deterministic sampling to select submissions, questions, and overlapping auditors.
4. Deliver the selected benchmark evidence and inclusion proofs only to the assigned auditors. Each recipient gets a separately encrypted envelope. Enforce authenticated key registration, expiry, replay state, and delivery deadlines.

5. Auditors commit answers before seeing peers' answers. Deliver openings and evidence privately to authorized reviewers. Resolve disagreement under the declared benchmark review policy before final weights.

The live policy must fix the sample size and auditor quorum. For pilot measurement, use three independently assigned auditors on each sampled task and retain all disagreements; stake identities alone do not establish operator independence. Question checks supplement whole-revision review rather than certifying every unexamined claim.

If b of N committed items contain detectable defects and s items are sampled uniformly without replacement, the probability of sampling at least one defective item is

$$1 - \frac{\binom{N-b}{s}}{\binom{N}{s}}. \tag{5}$$

Actual detection is weaker when questions miss the defect or reviewers fail to recognize it. This makes audit coverage measurable without claiming that sampling proves full fidelity.

Pin drand's schedule instead of assuming the chain's last stored pulse is the latest public pulse. A commitment finalized after a round becomes public cannot use that round as unknown future randomness. Predeclare wait or abort behavior for missed deadlines and unavailable beacons; do not select a favorable replacement round. drand specification [32], on-chain randomness pitfalls [33].

Subtensor's timelocked commit-reveal protects validator weights from immediate copying. It is separate from audit selection and recipient encryption. Weight plaintext eventually becomes public; customer inputs must not be encrypted with that eventual-reveal mechanism. Use the supported SDK/runtime path rather than hardcoding a CRv4 schedule. Bittensor commit-reveal [34].

10 Trust assumptions and adversarial behavior

The chain can enforce authenticated accounting and commitment consistency. Recipient encryption limits who can decrypt protocol deliveries. A verified beacon permits reproducible selection after the commitment deadline. Each guarantee has a narrower scope than useful, confidential writing inference.

Risk	Proposed response and remaining assumption
Customer text reaches another miner or gateway	Encrypt locally to the pinned assignment; no alternate keys or automatic reassignment
Assigned miner leaks decrypted text	Minimize retention and enforce service terms; endpoint nondisclosure is not cryptographically proved
Miner retains private weights but fabricates quality	Score benchmark outputs against hidden labels and independent reviewers
Miner copies reference content or removes difficult claims	Full preservation review, copying checks, and explicit exploit controls
Miner's private search and retry costs are unverifiable	Miners fund their own attempts; score committed reports under fixed deadlines; do not reimburse claimed private costs
Validators collude or score inconsistently	Random overlapping assignments, independent commitments, retained disagreements; honest review remains necessary
Selected subnet detector favors a transformation miner	Prior-round qualification, independent opponent assignment, signed responses and blinded controls; private execution and operator independence remain assumptions
Pangram changes or serves conflicting records	Pin the reported interface/version, retain observations, and suspend affected comparisons under a fixed policy
Customer consumes work and withholds payment	Bounded pilot exposure and explicit miner nonpayment risk; private fair exchange remains unresolved
Miner manufactures paid demand	Keep customer revenue separate from emission scoring

The benchmark also needs private source rotation, duplicate detection, and delayed release of retired examples. A service may recognize a repeated test or outsource a benchmark response; opaque version labels do not attest execution. Changing the benchmark or judge after observing a miner's answer would invalidate the declared comparison. Publish protocol versions and aggregate outcomes while keeping customer material and active evidence private.

11 Existing evidence and implementation status

The current Rust representation uses word occurrences and grammatical features from spaCy annotations. It scores complete sparse distributions and learns selected coordinate corrections. It does not require an LLM to enumerate or measure deterministic edit candidates. The initial detector/author models and editing rules provide experimental tools, not a demonstrated end-to-end product.

Experiment	Result	Scope
Blog author retrieval	56.30% top-1	600 evaluation authors in six 100-author galleries; 1,206 queries
Global Voices transfer	73.81% combined; 77.98% lexical-only	Fourteen provisionally qualified authors; 58 queries; topic/editorial confounds
Compact document projection	32.82% Blog; 25.00% Global Voices	Development-selected linear encoder; previously evaluated test corpora
Remove unselected contributions from old scorer	40.83% Blog; 29.17% Global Voices	Conditional diagnostic retaining old pooling and calibration

The projection experiment shows why 3,557 learned coordinate weights must not be mistaken for a sufficient fixed input vocabulary. The full model also retains unselected word and grammar contributions. Neither author retrieval nor these feature ablations establish meaning-preserving editing or detector evasion. In a separate edit diagnostic, replacing *may* with *must* improved author proximity despite changing the claim. Independent preservation checks are therefore central to the design. Author projection study [35], edit counterexamples [36].

Implemented components include Rust tools for corpus collection and evaluation, portable author scoring, offline challenge and submission checks, a public Pangram report reader, and an HPKE prototype that encrypts receipts for assigned validators. The receipt prototype relies on caller-supplied trusted assignments; it is not a customer-request protocol or deployed key registry. Receipt implementation [37].

The project does not yet demonstrate a reliable sub-10% rewriting model, calibrated production-origin service, reviewed commercial training release, authenticated live benchmark ledger, customer escrow contract, or private serving endpoint. The results justify a measured pilot and constrain its design; they do not establish readiness for emissions or customer funds.

12 Development and launch criteria

First, run a local tournament with 24 rights-cleared tasks across the three interfaces. Evaluate detection on known-label author and origin tasks against fixed probabilistic baselines. For editing tasks, compare unchanged text, deterministic edits, and one general editor under the same briefs. Have two blinded readers assess preservation and preference, including intentional omissions, uncertainty changes, and padding. Freeze the scoring policy before comparison and publish aggregate agreement, coverage, cost, and failure cases.

In parallel, use synthetic jobs and test funds to verify assigned-miner encryption, customer-only result delivery, key substitution rejection, replay prevention, expiry, acknowledgment, and exactly one settlement. Exercise miner outage, wrong keys, unacknowledged consumed results, and attempted automatic reassignment. Neither a validator nor a gateway should have a decryption path. Measure encrypted payload sizes and chain costs before enabling on-chain transport. No real customer text is needed for that pilot.

Use the writer pilot to establish acquisition and review costs. Refit public starter models only on data cleared for the intended release. Evaluate author and origin models on new sources, then train editing adapters after enough reviewed pairs exist. Reward scaling requires demonstrated judge reliability and preservation performance, with uncertainty reported on unseen sources.

Before a live subnet launch, resolve the supported Pangram retrieval/billing integration, key and assignment registry, replay ledger, subnet-detector qualification and response verification, cross-evaluation assignments, audit quorum and sample sizes, empty-pool weight policy, and mechanism allocations. Before paid customer launch, separately establish acceptable nonpayment losses, retention terms, encrypted delivery reliability, and alpha escrow behavior under finality and timeouts. Verify quote expiry, capacity reservations, exact alpha accounting, release and refund permissions, and fee handling. These decisions must be concrete protocol parameters or measured acceptance criteria before funds depend on them.

The next deliverable is a small reproducible tournament and private-job pilot, followed by revised parameters based on observed failures and reader judgments. The whitepaper remains a design draft until those results support deployment.

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